

Support Vector Machine

Mahdi Roozbahani
Georgia Tech

Math moment

$$7 \times 0 = 0$$

$$3 \times 0 = 0$$

$$0 = 0$$

$$7 \times 0 = 3 \times 0$$

$$7 = 3$$



Outline

- Precursor: Linear Classifier and Perceptron 
- Support Vector Machine
- Parameter Learning

Binary Classification

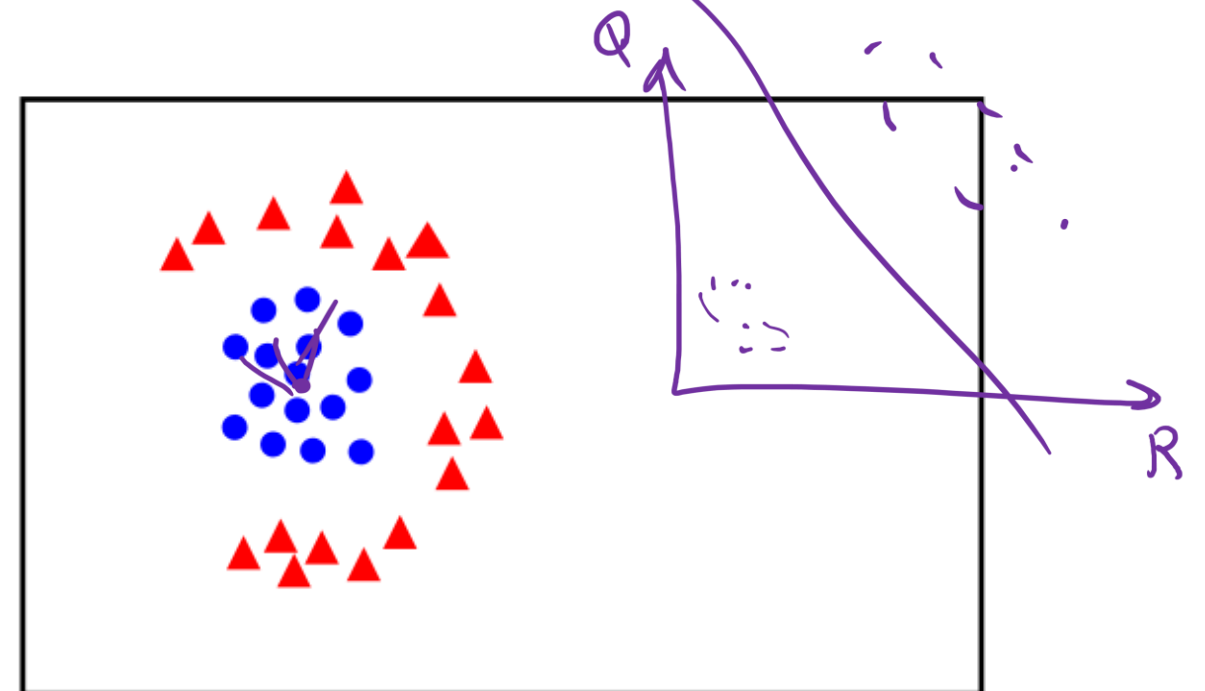
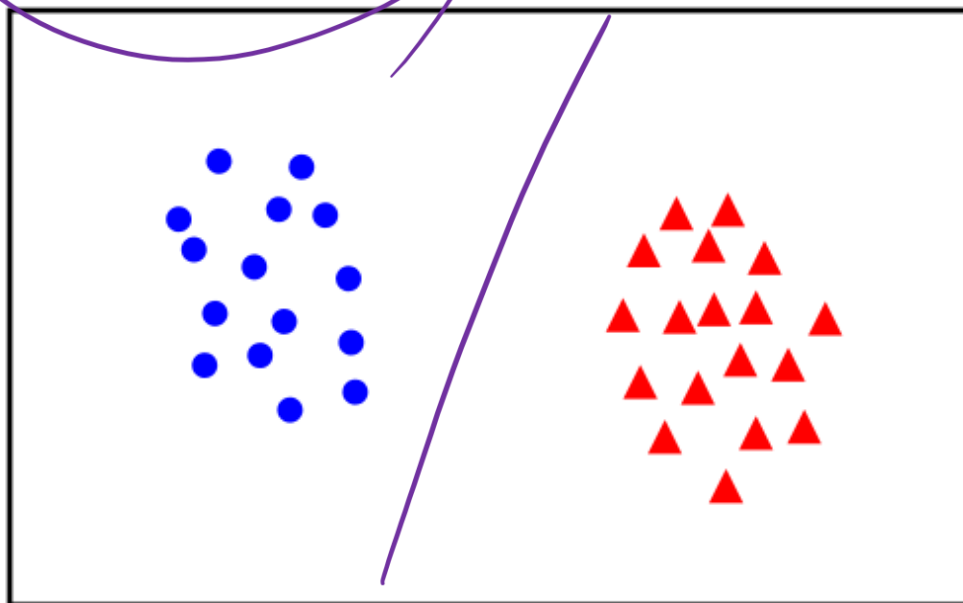
Given training data $(x^{(i)}, y^{(i)})$ for $i = 1 \dots N$, with $x^{(i)} \in \mathbb{R}^d$ and $y^{(i)} \in \{-1, 1\}$, learn a classifier $f(\mathbf{x})$ such that

$f(x^{(i)}) = \text{KF} = \theta_0 + \theta_1 x_1 + \dots + \theta_d x_d$

$$f(x^{(i)}) \begin{cases} \geq 0 \longrightarrow +1 \\ < 0 \longrightarrow -1 \end{cases}$$

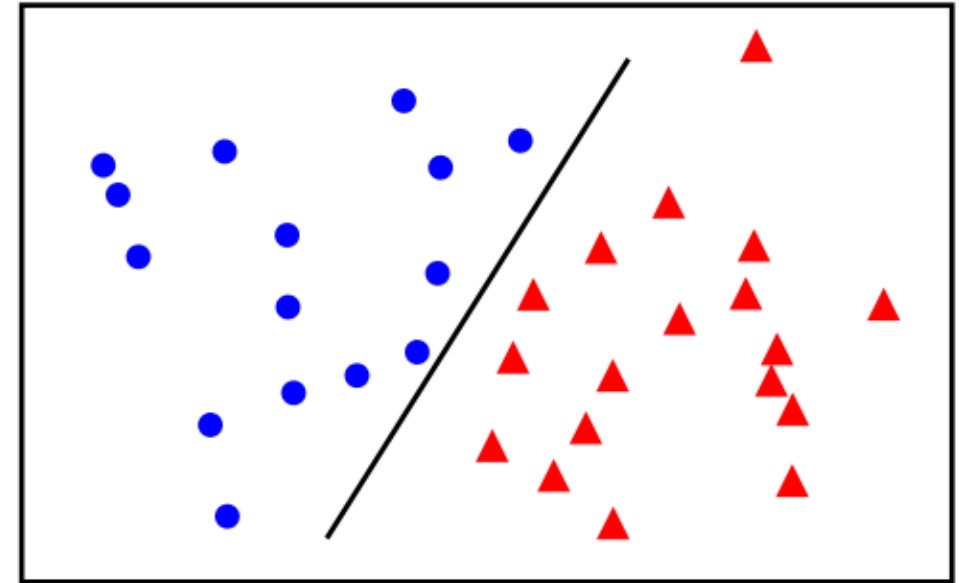
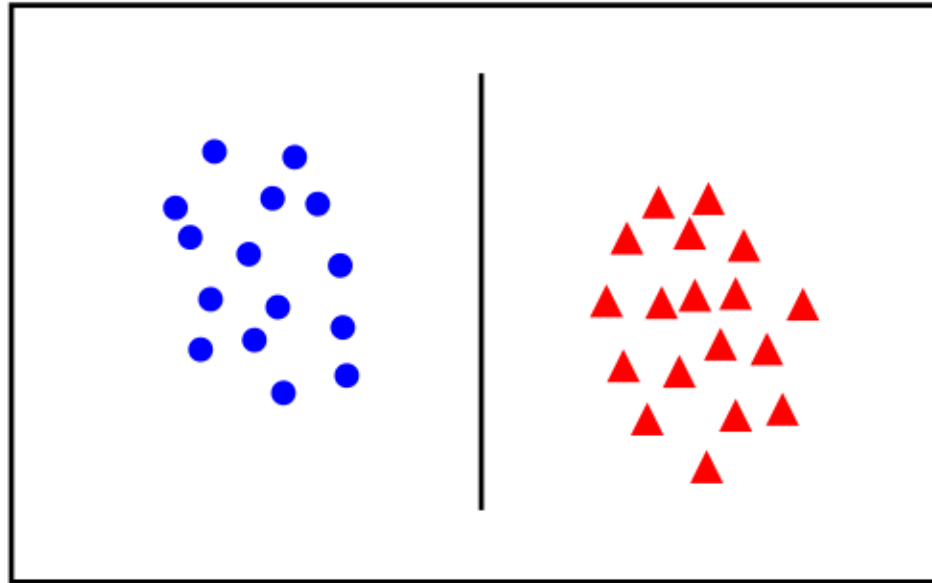
i.e. $y^{(i)} f(x^{(i)}) > 0$ for a correct classification.

$\begin{matrix} + & + \\ - & - \end{matrix}$
actual *predicted*

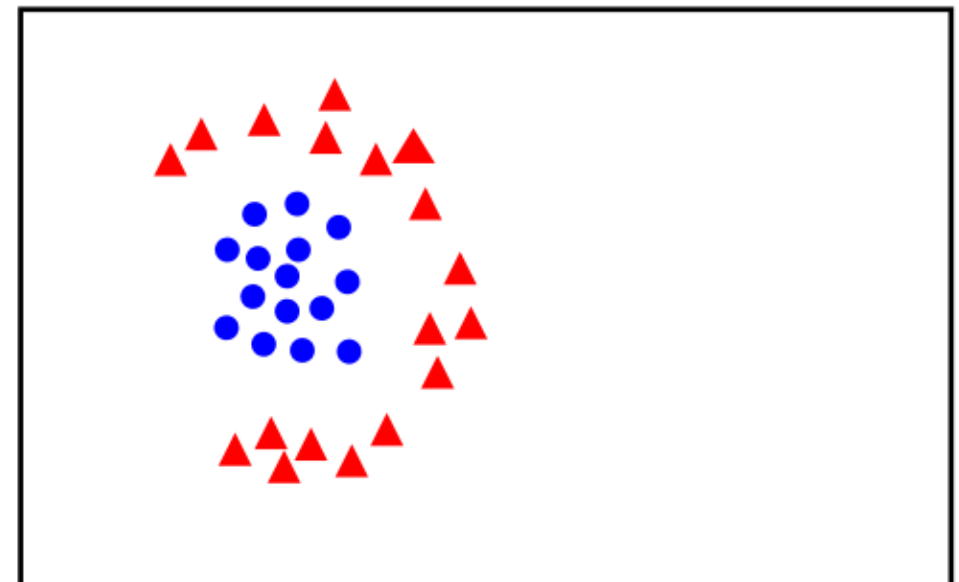
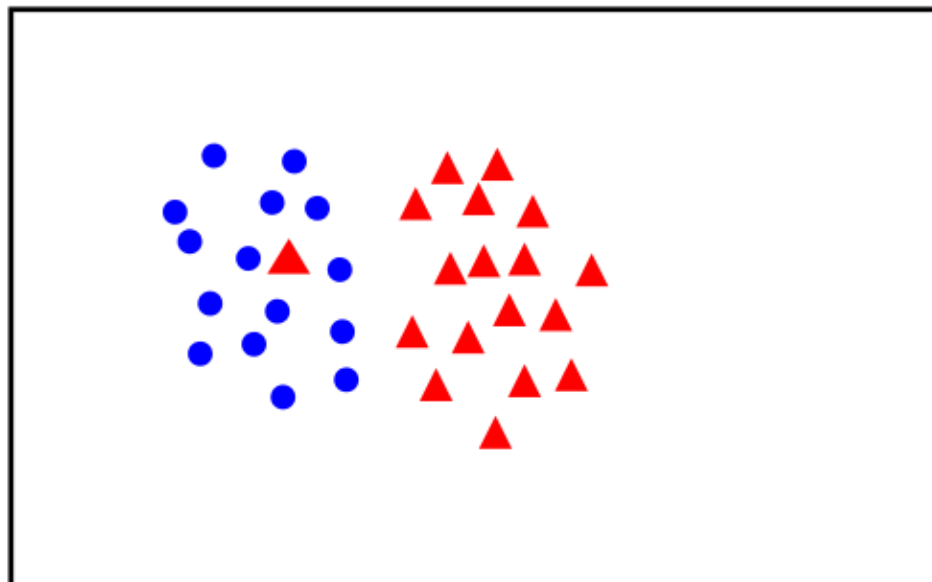


Linear Separability

linearly
separable



not
linearly
separable



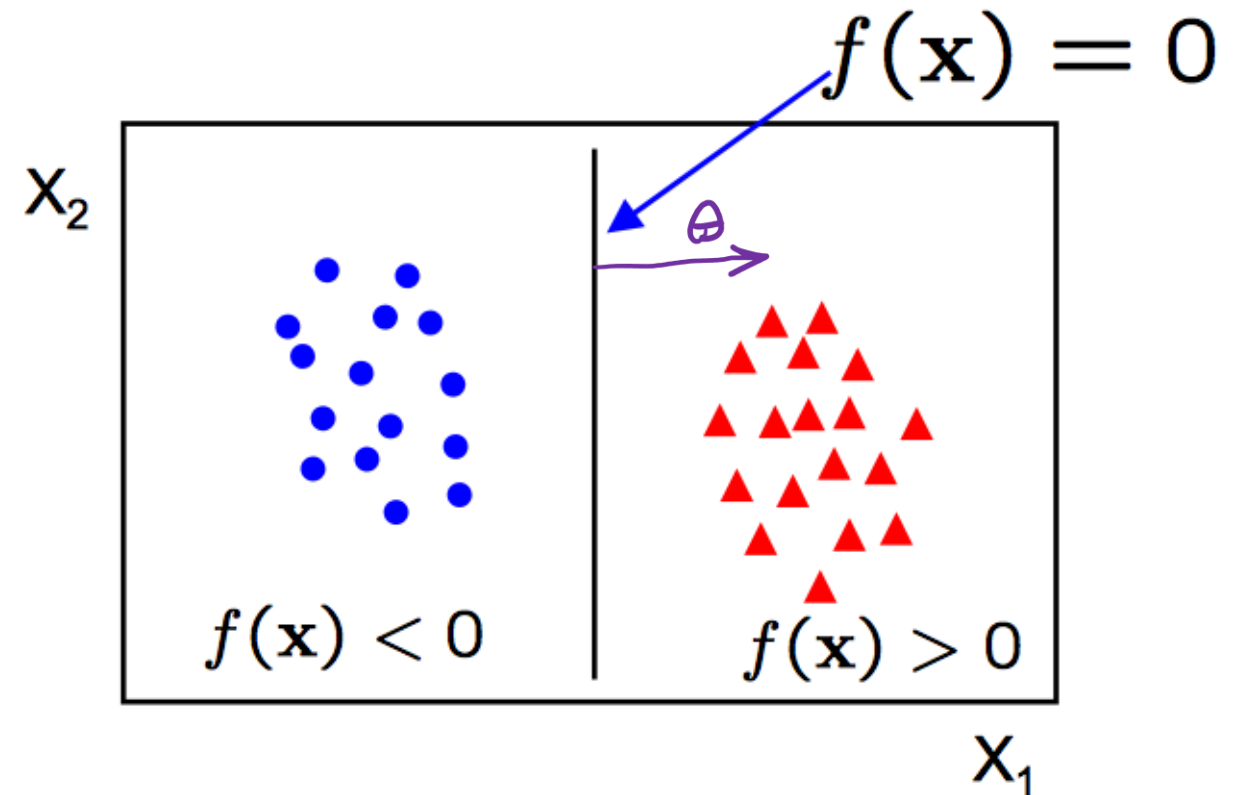
Linear Classifier

$$f(x) = x\theta = 0$$

vector vector

A linear classifier has the form

$$f(x) = x\theta + \theta_0$$

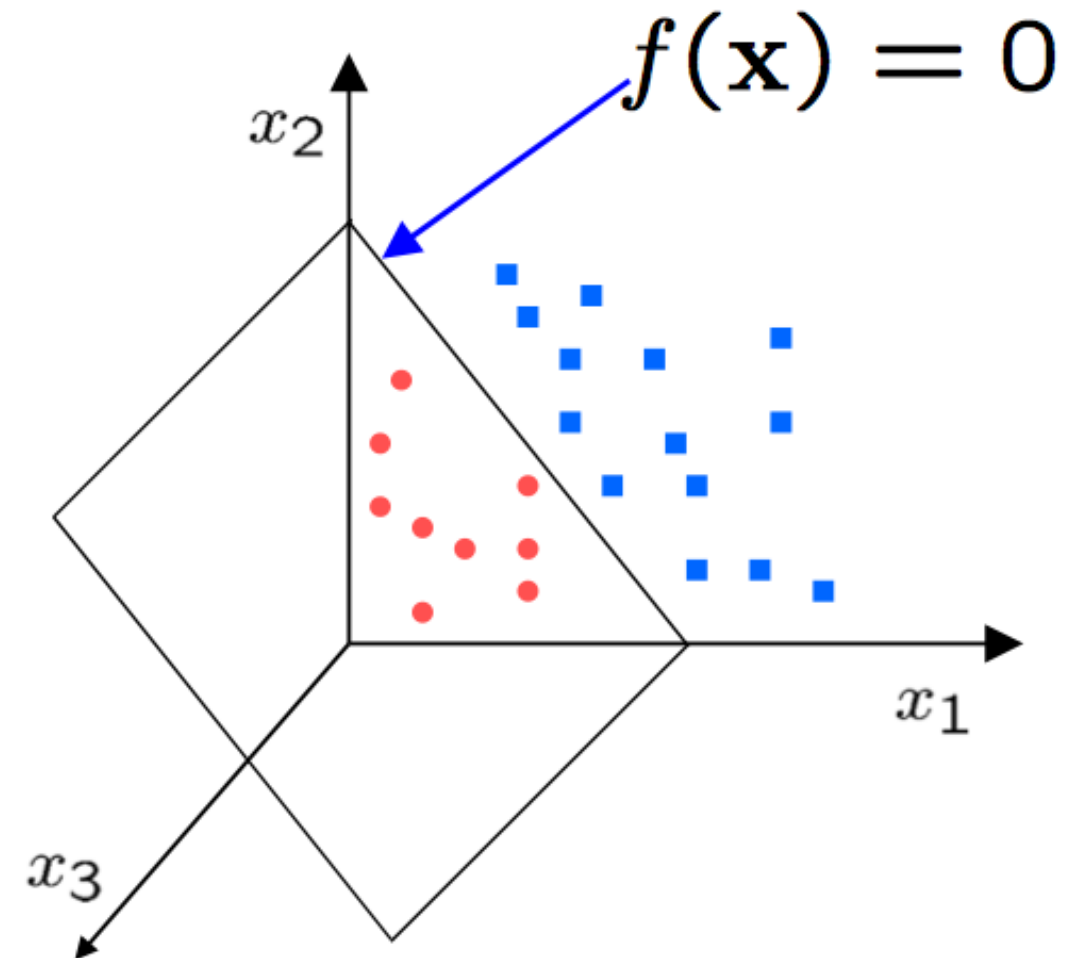


- in 2D the discriminant is a line
- θ is the **normal** to the line, and θ_0 the **bias**
- θ is known as the **weight vector**

Linear Classifier (higher dimension)

A linear classifier has the form

$$f(x) = x\theta + \theta_0$$



- in 3D the discriminant is a plane, and in nD it is a hyperplane

The Perceptron Classifier

Considering $\mathbf{x}$ is linearly separable and $\mathbf{y}$ has two labels of $\{-1, 1\}$

$$f(x^{(i)}) = x^{(i)} \theta \quad \text{Bias is inside } \theta \text{ now}$$

How can we separate datapoints with label 1 from datapoints with label -1 using a line?

Perceptron Algorithm:

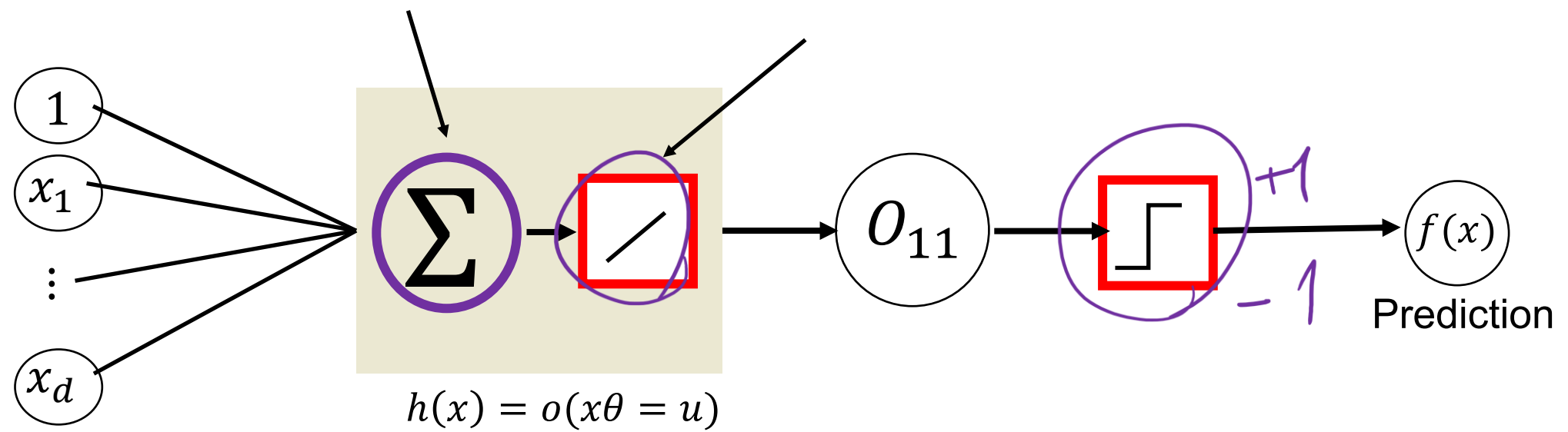
- Initialize $\theta = 0$
- Go through each datapoint $\{x^{(i)}, y^{(i)}\}$
 - If $x^{(i)}$ is misclassified then $\theta^{t+1} \leftarrow \theta^t + \alpha y^{(i)} x^{(i)}$
- Until all datapoints are correctly classified

Misclassified

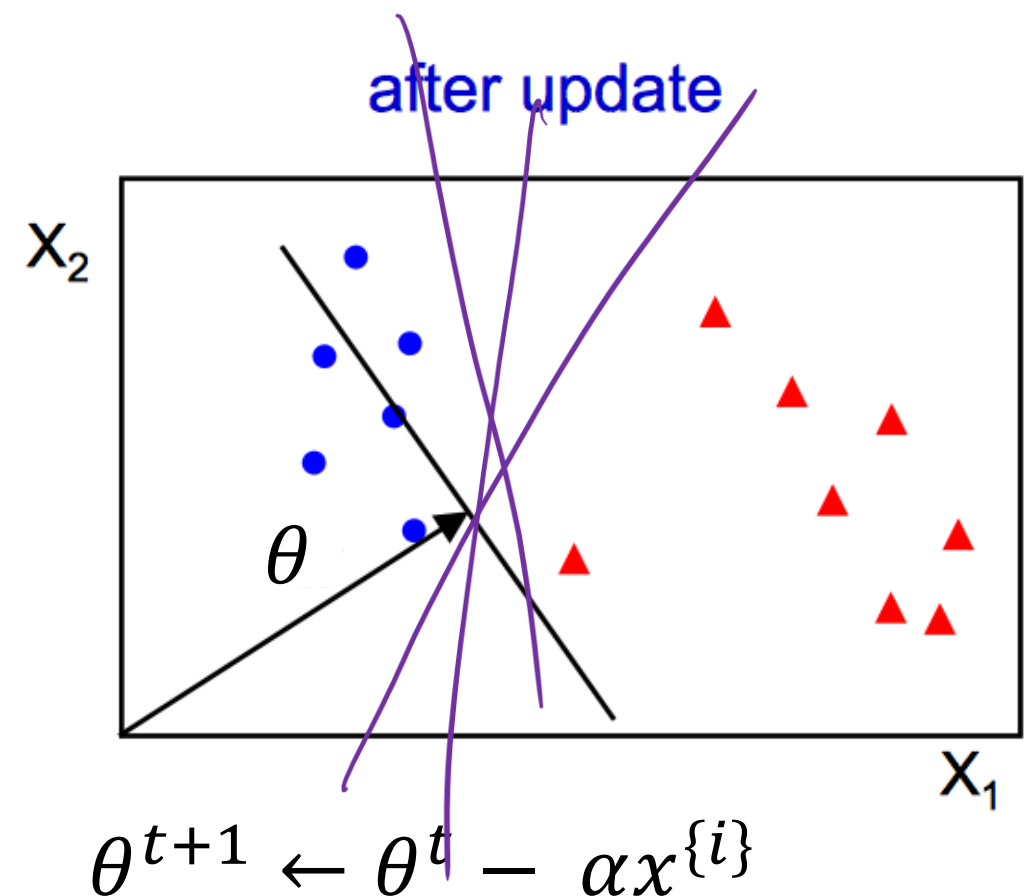
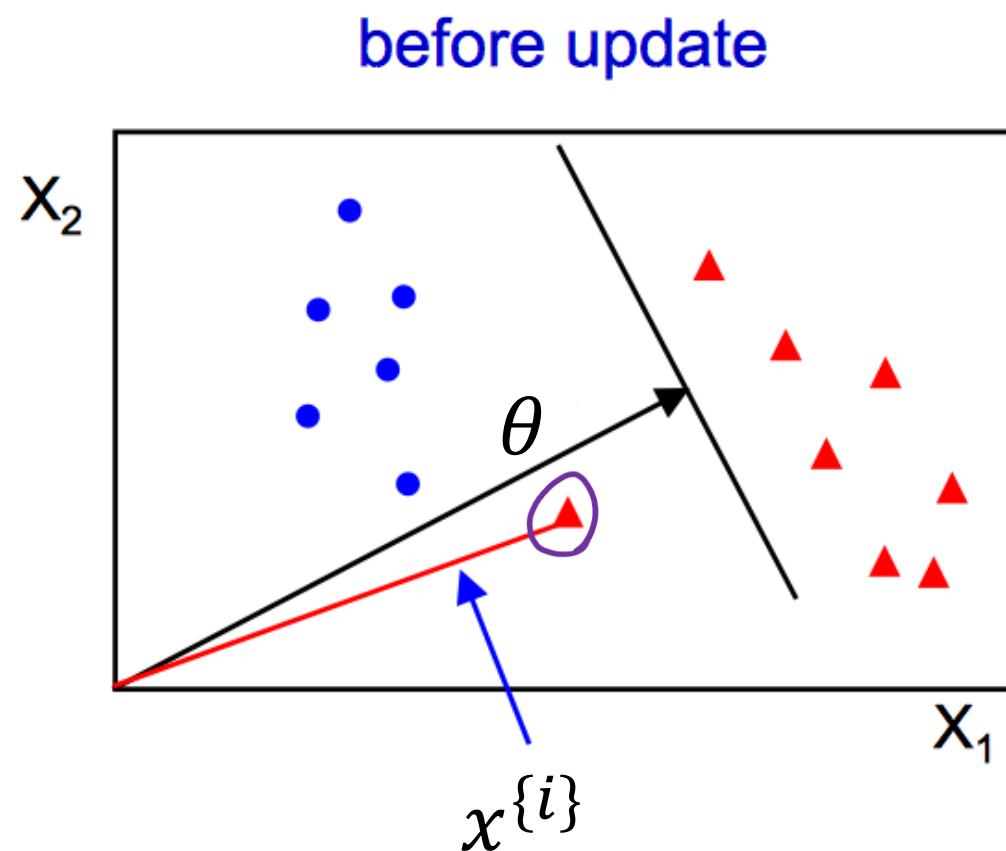
$$\text{Ex. } y^{(i)} f(x^{(i)}) < 0$$

actual predicted

Summation



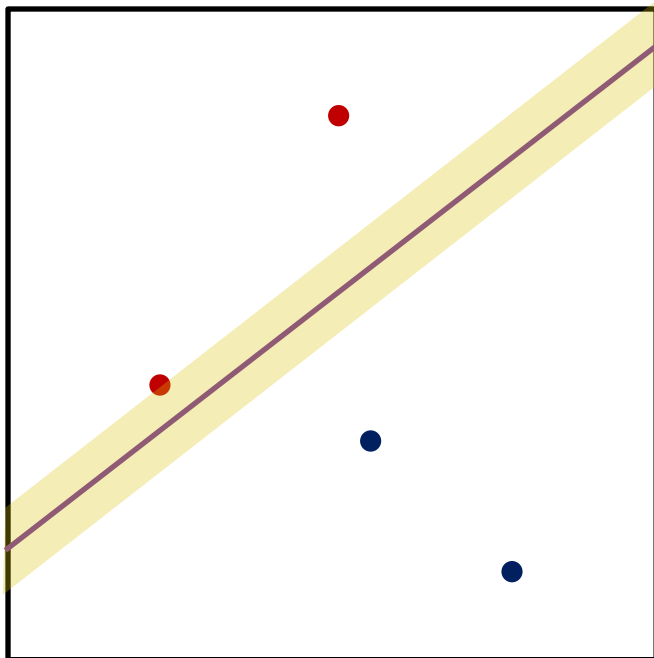
- Initialize $\theta = 0$
- Go through each datapoint $\{x^{(i)}, y^{(i)}\}$
 - If $x^{(i)}$ is misclassified then $\theta^{t+1} \leftarrow \theta^t + \alpha y^{(i)} x^{(i)}$
- Until all datapoints are correctly classified



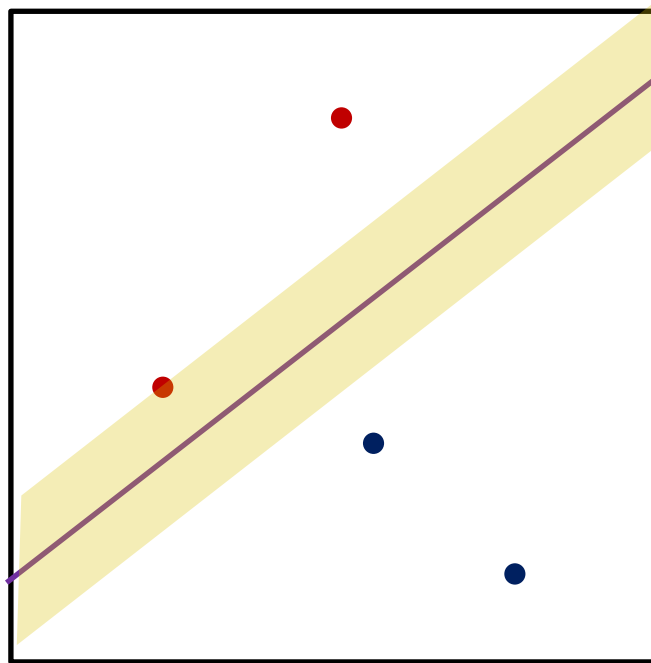
Linear separation

We can have different separating lines

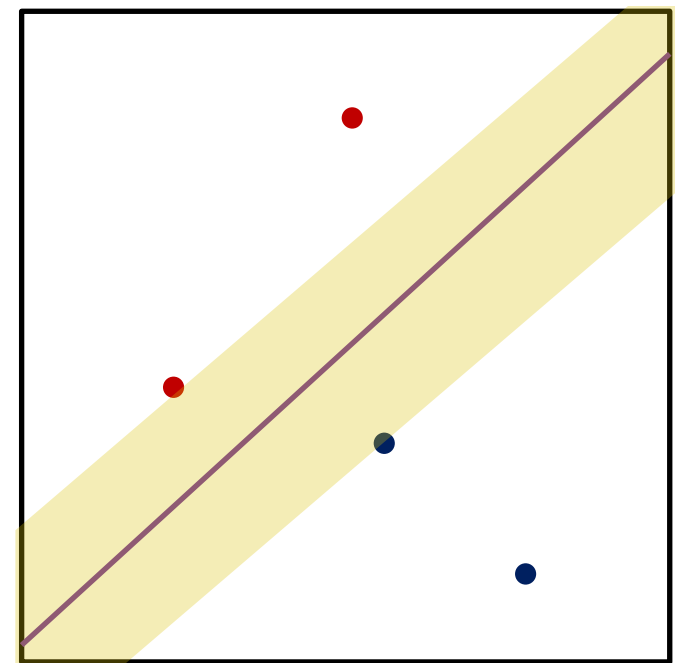
①



②



③



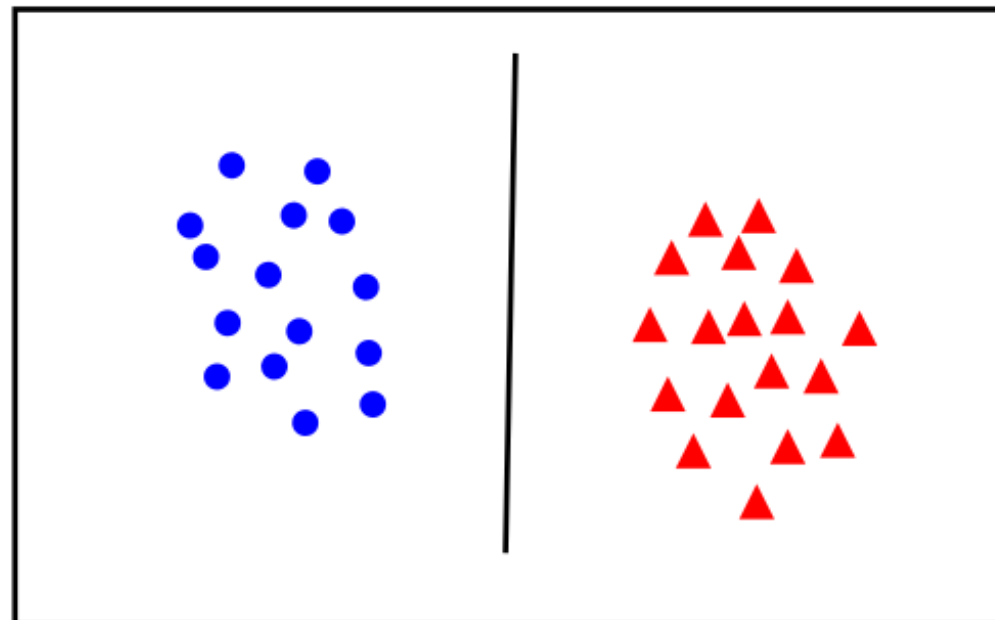
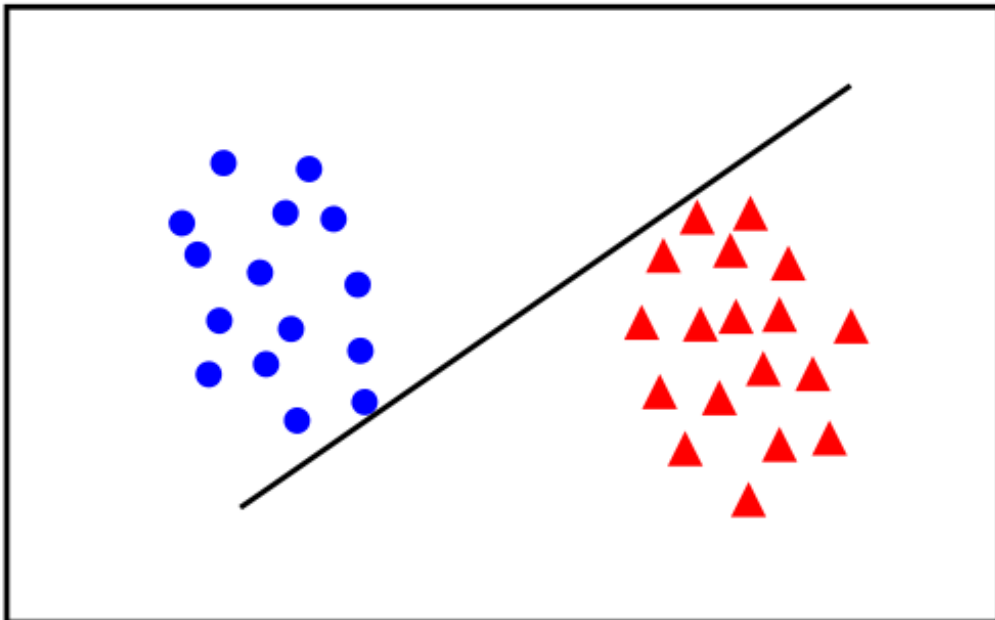
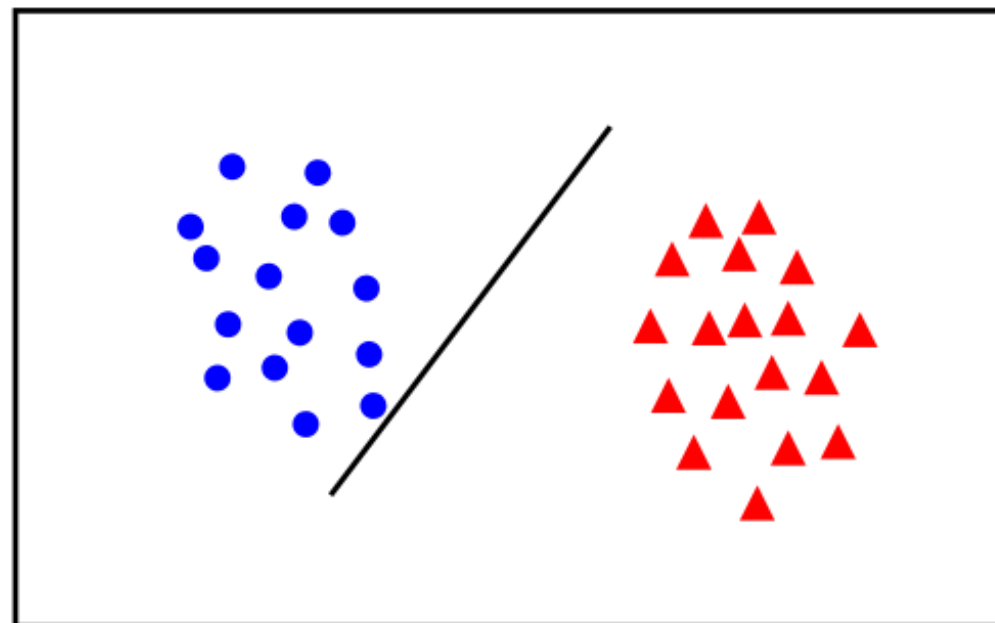
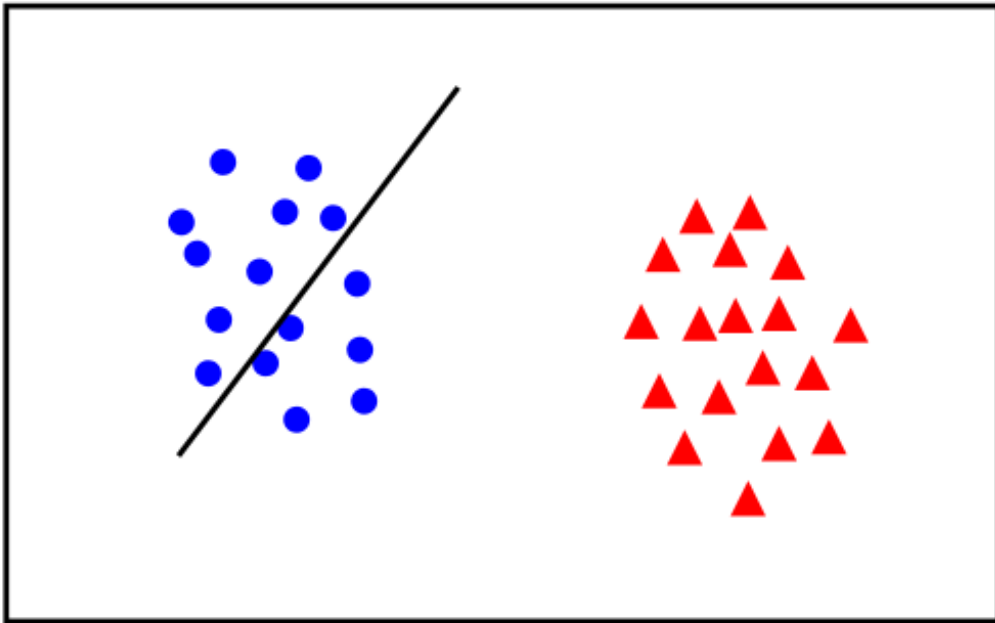
Which line is the best?

All cases, error is zero and they are linear, so they are all good for generalization.

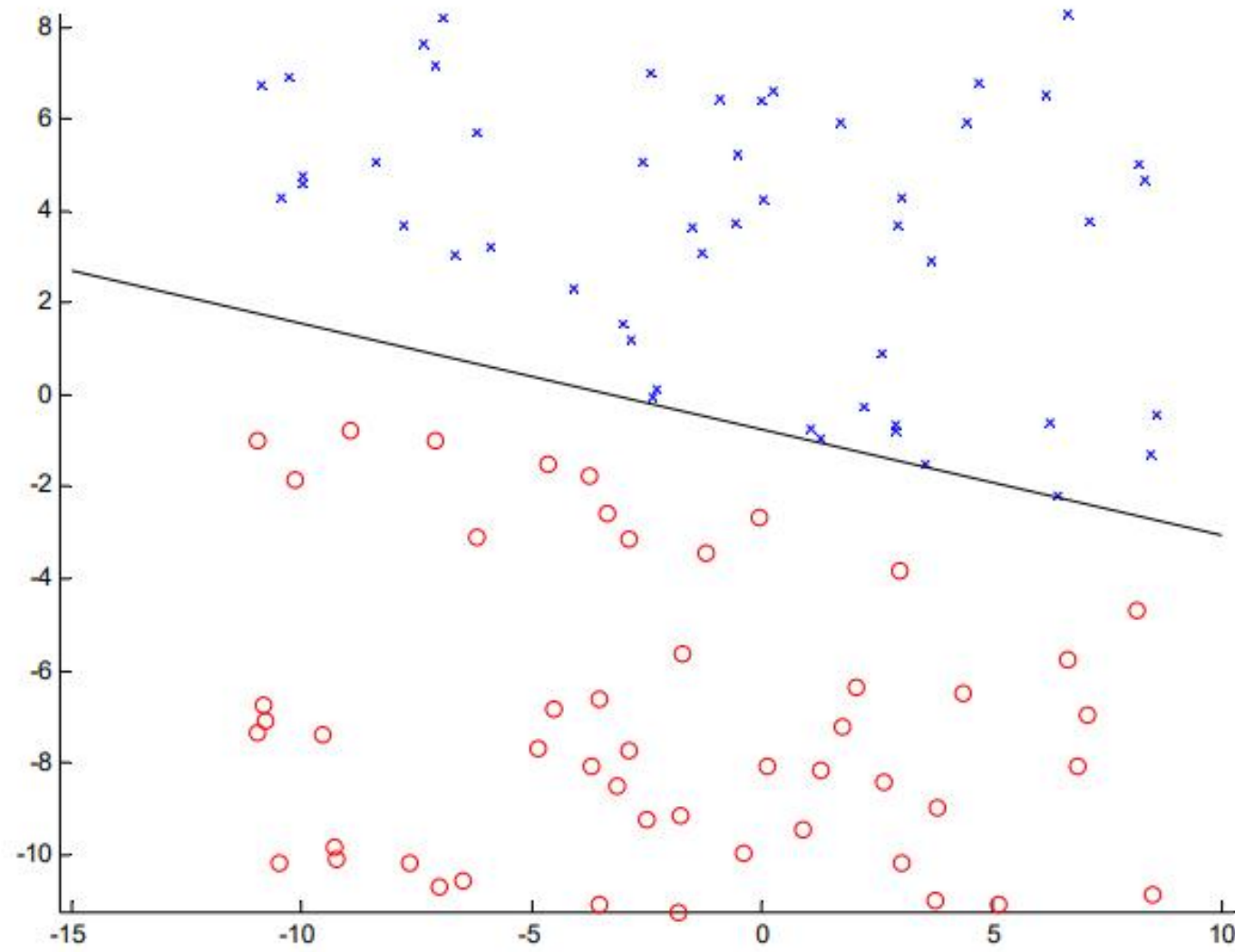
Why is the bigger margin better?

What θ maximizes the margin?

What is the Best θ ?



Perceptron example



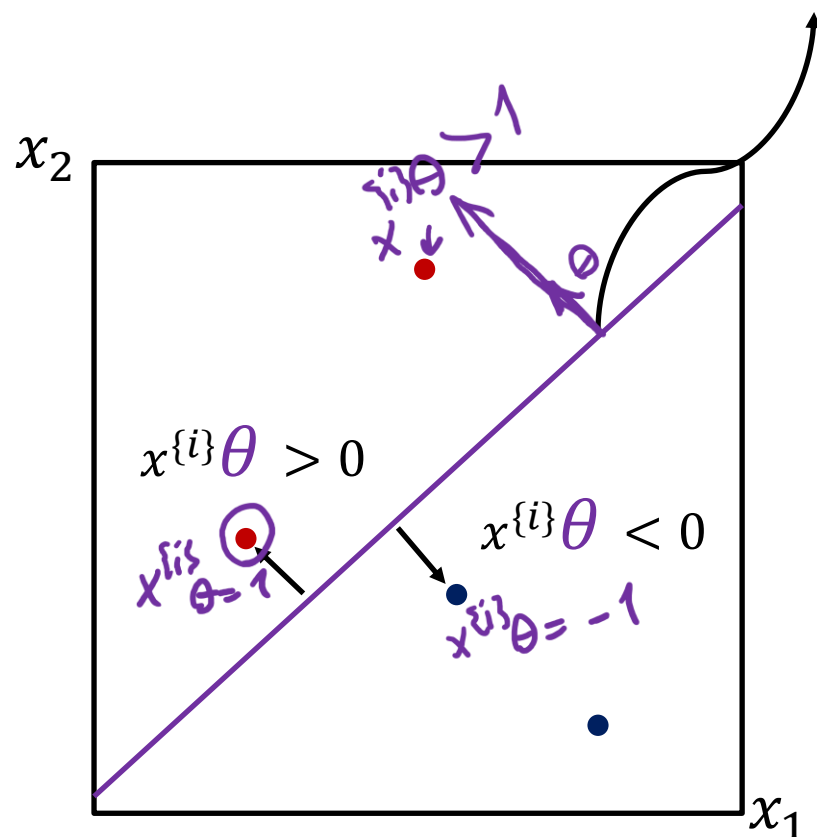
- if the data is linearly separable, then the algorithm will converge
- convergence can be slow ...
- separating line close to training data
- we would prefer a larger margin for generalization (better generalization)

Outline

- Precursor: Linear Classifier and Perceptron
- Support Vector Machine 
- Parameter Learning 

Finding θ with a **fat** margin

Solution (decision boundary) of the line: $x\theta = 0$



Let x^{i} to be the nearest data point to the line (plane):

$$|x^{i}\theta| > 0$$

Our line solution is $x\theta = 0$

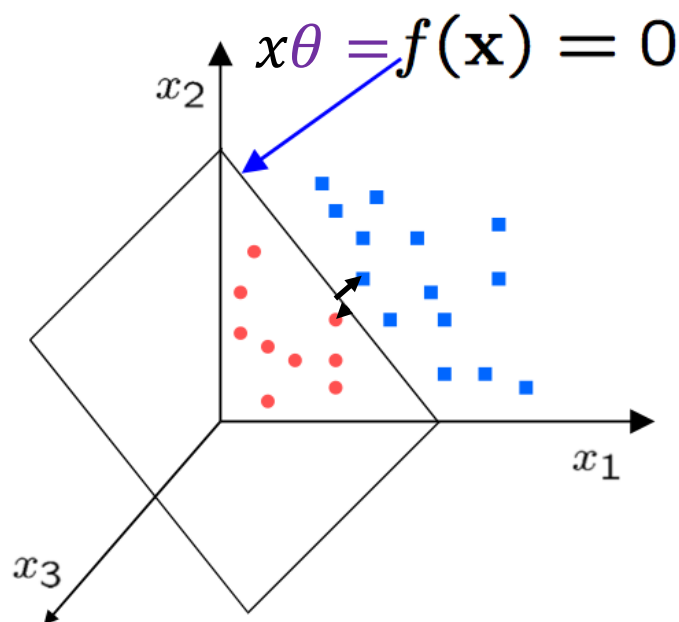
Does it matter if I scale up or down θ for the decision boundary?

LCF

$$|x^{i}\theta| = 1 \rightarrow \text{normalization}$$

Let's pull out θ_0 from $\theta = (\theta_1, \dots, \theta_d)$ and call it be b

Decision boundary would be: $x\theta + b = 0$



Computing the distance

The distance between $x^{\{i\}}$ and the line $x\theta + b = 0$ where $|x^{\{i\}}\theta + b| = 1$

The vector θ is perpendicular to the decision line.

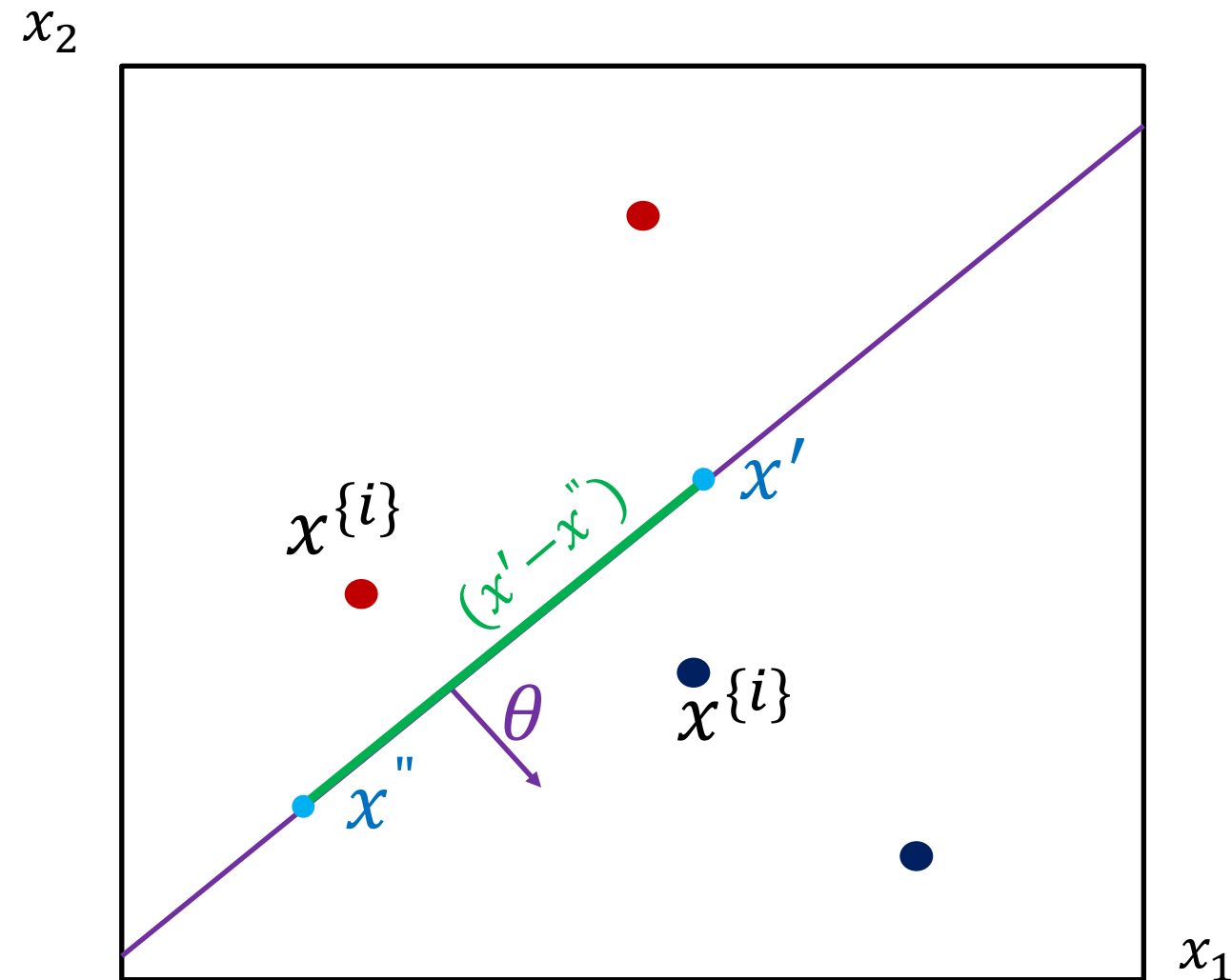
You should ask me why?

Consider x' and x'' on the plane

$$x'\theta + b = 0 \quad \text{and} \quad x''\theta + b = 0$$

$$\Downarrow$$
$$x'\theta + b = x''\theta + b$$

$$\Rightarrow (x' - x'')\theta = 0$$



What is the distance of my fat margin?

What is the distance between $x^{\{i\}}$ and the plane?

Let's take any point x on the line:

Distance would be projection of $(x^{\{i\}} - x)$ vector on θ .

To project the vector, we need to normalize θ to get the unit vector.

$\hat{\theta} = \frac{\theta}{\|\theta\|} \Rightarrow \text{distance} = |(x^{\{i\}} - x) \hat{\theta}|$ which is the dot product

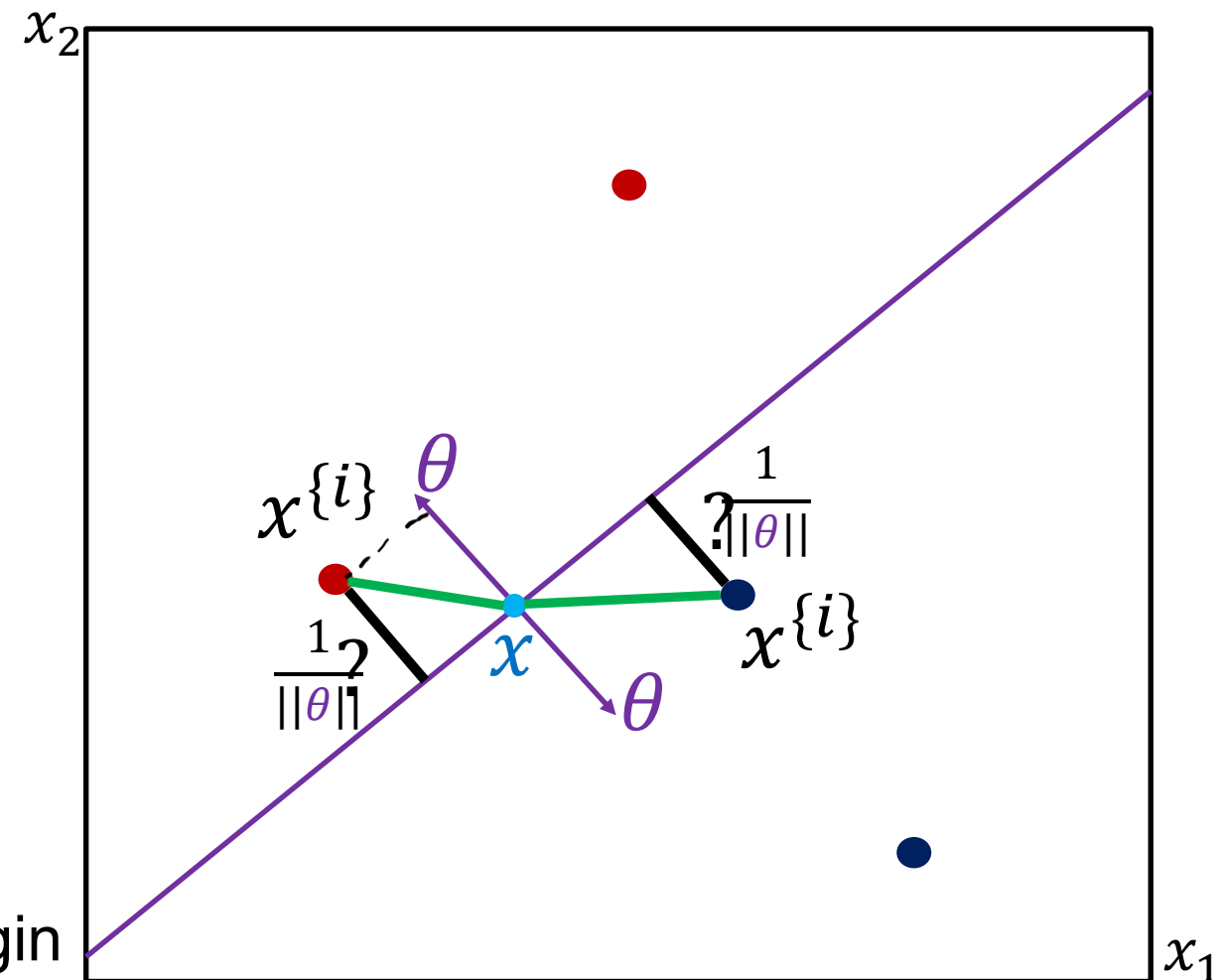
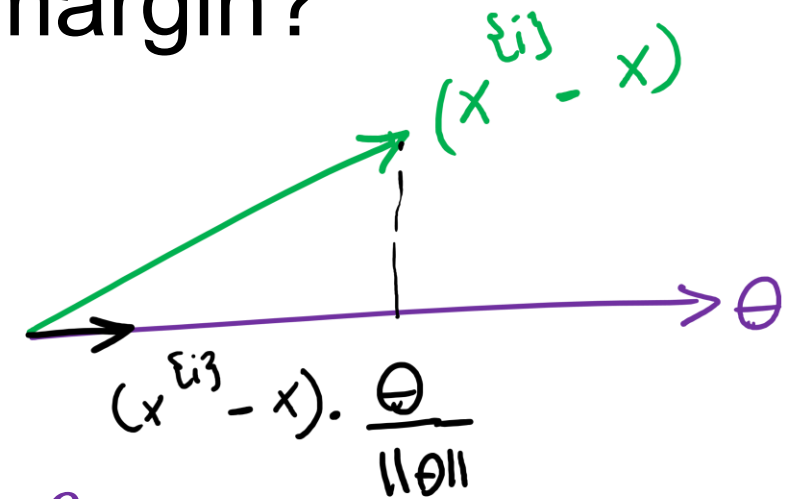
$$\text{distance} = \frac{1}{\|\theta\|} |(x^{\{i\}} \theta - x \theta)|$$

$$= \frac{1}{\|\theta\|} | \underbrace{(x^{\{i\}} \theta + b)}_1 - \underbrace{(x \theta - b)}_0 | = \frac{1}{\|\theta\|}$$

My constraint
 $|x^{\{i\}} \theta + b| = 1$

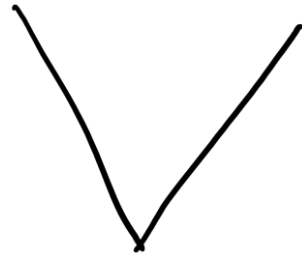
A point on the
decision line
 $x \theta + b = 0$

The margin



Now we need to maximize the margin

Maximize $\frac{1}{||\theta||}$



Subject to Min value of $|x^{\{i\}}\theta + b| = 1 \Rightarrow \text{nearest neighbour}$
 $i = 1, 2, \dots, N$

There is a “min” in our constraining; it can be hard to optimize this problem(non-convex form)

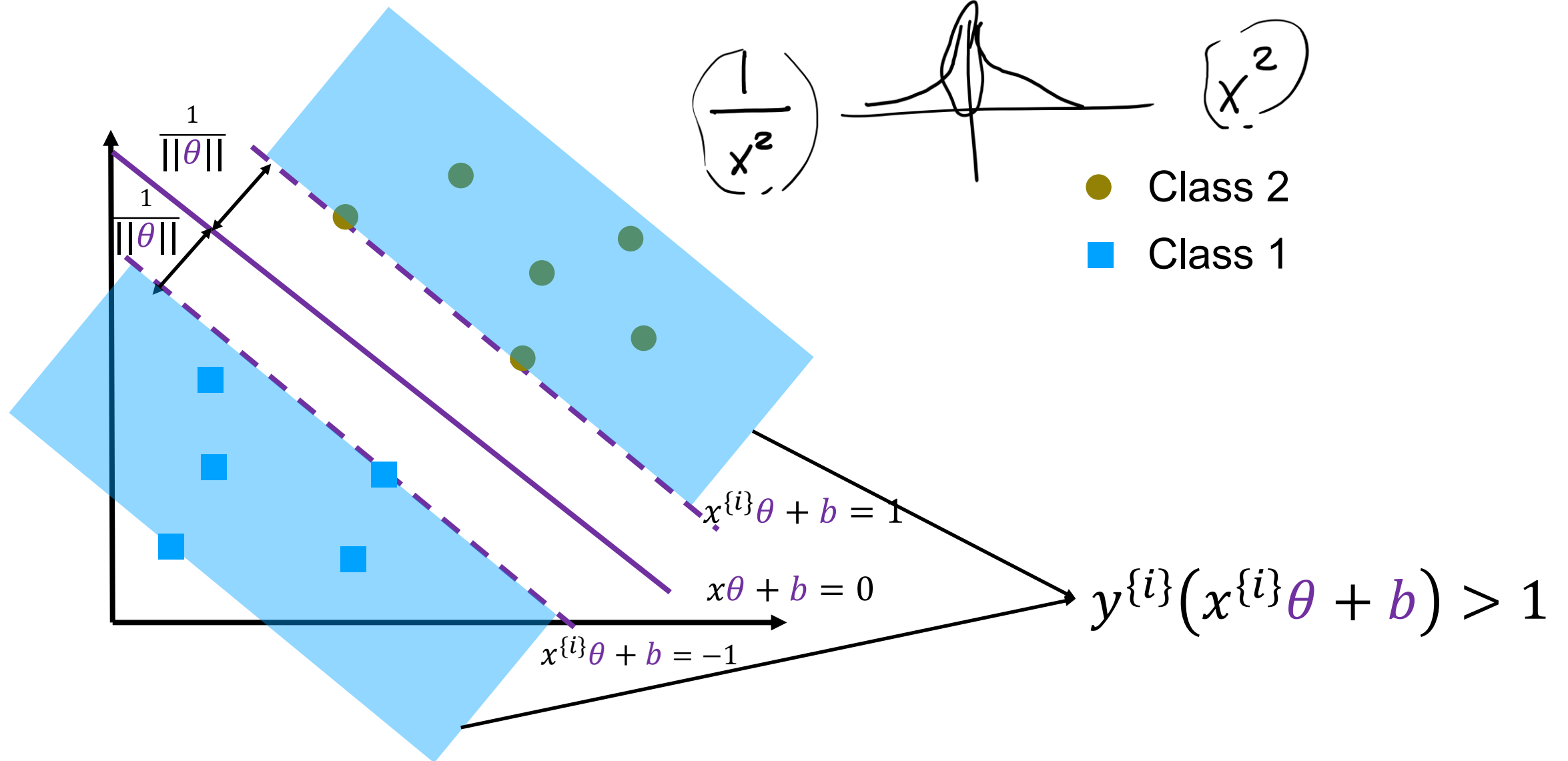
Can I write the following term to get rid of absolute value?

$$|x^{\{i\}}\theta + b| = y^{\{i\}}(x^{\{i\}}\theta + b) \Rightarrow \text{for a correct classification}$$

$$\text{If min } |x^{\{i\}}\theta + b| = 1 \Rightarrow \text{so it can be at least 1}$$

Maximize $\frac{1}{||\theta||}$

Subject to $y^{\{i\}}(x^{\{i\}}\theta + b) \geq 1$ for $i = 1, 2, \dots, N$



Maximize $\frac{2}{\|\theta\|}$

If $\theta \neq 0$, there exists a max value

Subject to $y^{\{i\}}(x^{\{i\}}\theta + b) \geq 1$ for $i = 1, 2, \dots, N$

$\theta\theta^T = \|\theta\|^2$

Minimize $\frac{1}{2}\theta\theta^T$

Subject to $y^{\{i\}}(x^{\{i\}}\theta + b) \geq 1$ for $i = 1, 2, \dots, N$

Constrained optimization

$$\text{Minimize } \frac{1}{2} \theta \theta^T$$

$$\text{Subject to } y^{\{i\}}(x^{\{i\}}\theta + b) \geq 1 \text{ for } i = 1, 2, \dots, N$$

$$\theta \in \mathbb{R}^d, b \in \mathbb{R}$$

Using Lagrange method: But wait, there is an **inequality** in our constraints

We use Karush-Kuhn-Tucker (KKT) condition to deal with this problem

$$g(x) = y^{\{i\}}(x^{\{i\}}\theta + b) - 1 \quad \alpha = \text{lagrange multiplier}$$

We need to optimize

- 1) Stationary $\theta, b, \text{ and } \alpha$
- 2) $g(x) \geq 0$ Primal feasibility
- 3) $\alpha \geq 0$ Dual feasibility
- 4) $g(x)\alpha = 0$ Complementary slackness $\Rightarrow \begin{cases} g(x) > 0, & \alpha = 0 \\ \alpha > 0, & g(x) = 0 \end{cases}$

Lagrange formulation

$$\text{Minimize } \frac{1}{2} \theta \theta^T \quad \text{s.t.} \quad y^{\{i\}}(x^{\{i\}} \theta + b) - 1 \geq 0$$

$$\mathcal{L}(\theta, b, \alpha) = \frac{1}{2} \theta \theta^T - \sum_{i=1}^N \alpha^{\{i\}} (y^{\{i\}}(x^{\{i\}} \theta + b) - 1)$$

Minimize w.r.t θ and b and maximize w.r.t each $\alpha^{\{i\}} \geq 0$

$$\theta = \sum_{i=1}^N \alpha^{\{i\}} y^{\{i\}} x^{\{i\}}$$

$$\nabla_{\theta} \mathcal{L}(\theta, b, \alpha) = \theta - \sum_{i=1}^N \alpha^{\{i\}} y^{\{i\}} x^{\{i\}} = 0$$

$$\nabla_b \mathcal{L}(\theta, b, \alpha) = - \sum_{i=1}^N \alpha^{\{i\}} y^{\{i\}} = 0$$

$$\theta = \sum_{i=1}^N \alpha^{(i)} y^{(i)} x^{(i)}$$

$$\sum_{i=1}^N \alpha^{(i)} y^{(i)} = 0$$

$$\sum \alpha^{(i)} y^{(i)} x^{(i)} \theta + b \sum \alpha^{(i)} y^{(i)}$$

Let's substitute these in the Lagrangian:

$$\mathcal{L}(\theta, b, \alpha) = \frac{1}{2} \theta \theta^T - \sum_{i=1}^N \alpha^{(i)} (y^{(i)} (x^{(i)} \theta + b) - 1)$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^N \alpha^{(i)} + \frac{1}{2} \theta \theta^T - \sum_{i=1}^N \alpha^{(i)} (y^{(i)} (x^{(i)} \theta + b))$$

$$\begin{aligned} \mathcal{L}(\theta, b, \alpha) &= \sum_{i=1}^N \alpha^{(i)} + \frac{1}{2} \theta \theta^T - \sum_{i=1}^N \alpha^{(i)} (y^{(i)} (x^{(i)} \theta)) - \sum_{i=1}^N \alpha^{(i)} + \frac{1}{2} \theta \theta^T - \theta \theta^T = \\ &= \sum_{i=1}^N \alpha^{(i)} - \frac{1}{2} \theta \theta^T \end{aligned}$$

primal form

$$\theta = \sum_{i=1}^N \alpha^{\{i\}} y^{\{i\}} x^{\{i\}}$$

$$\sum_{i=1}^N \alpha^{\{i\}} y^{\{i\}} = 0$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^N \alpha^{\{i\}} - \frac{1}{2} \theta \theta^T$$

$$\mathcal{L}(\theta, b, \alpha) = \sum_{i=1}^N \alpha^{\{i\}} - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{\{i\}} y^{\{j\}} \alpha^{\{i\}} \alpha^{\{j\}} x^{\{i\}} x^{\{j\}^T}$$

maximize *w.r.t* each $\alpha^{\{i\}} \geq 0$ for $i = 1, \dots, N$

and

$$\sum_{i=1}^N \alpha^{\{i\}} y^{\{i\}} = 0$$

The solution – quadratic programming

$$\max_{\alpha} \sum_{i=1}^N \alpha^{(i)} - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{(i)} y^{(j)} \alpha^{(i)} \alpha^{(j)} x^{(i)} x^{(j)T}$$

Quadratic programming packages usually use “min”

$$\min_{\alpha} \left[\frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{(i)} y^{(j)} \alpha^{(i)} \alpha^{(j)} x^{(i)} x^{(j)T} - \sum_{i=1}^N \alpha^{(i)} \right]$$

$$\alpha = \begin{bmatrix} \alpha^{(1)} \\ \alpha^{(2)} \\ \vdots \\ \alpha^{(N)} \end{bmatrix}_{N \times 1}$$

$$\min_{\alpha} \frac{1}{2} \alpha^T \begin{bmatrix} \underbrace{y^{(1)} y^{(1)}}_{1 \times 1} \underbrace{y^{(1)} y^{(1)}}_{1 \times 1} \underbrace{x^{(1)} x^{(1)T}}_{1 \times d} \underbrace{x^{(1)} x^{(1)T}}_{d \times 1} & \dots & y^{(1)} y^{(N)} x^{(1)} x^{(N)T} \\ \vdots & \ddots & \vdots \\ y^{(N)} y^{(1)} x^{(N)} x^{(1)T} & \dots & y^{(N)} y^{(N)} x^{(N)} x^{(N)T} \end{bmatrix} \alpha + (-I^T) \alpha$$

$\alpha^{(1)} + \alpha^{(2)} + \dots + \alpha^{(N)}$
 $\in \mathbb{R}_M$

$N \times 1$
 $N \times N$
 1×1

$$\min_{\alpha} \frac{1}{2} \alpha^T \begin{bmatrix} y^{\{1\}} y^{\{1\}} x^{\{1\}} x^{\{1\}T} & \dots & y^{\{1\}} y^{\{N\}} x^{\{1\}} x^{\{N\}T} \\ \vdots & \ddots & \vdots \\ y^{\{N\}} y^{\{1\}} x^{\{N\}} x^{\{1\}T} & \dots & y^{\{N\}} y^{\{N\}} x^{\{N\}} x^{\{N\}T} \end{bmatrix} \alpha + \underbrace{(-I^T) \alpha}_{\text{Linear term}}$$

Quadratic coefficients

Subject to $\sum_{i=1}^N \alpha^{\{i\}} y^{\{i\}} = y^T \alpha = 0$

Linear equality constraint

Pass these to a quadratic programming package

$$\text{lower bound}(0) \leq \alpha \leq \text{upper bound}(\infty)$$

$$\min_{\alpha} \frac{1}{2} \alpha^T Q \alpha - 1^T \alpha \quad \text{subject to} \quad y^T \alpha = 0; \alpha \geq 0$$

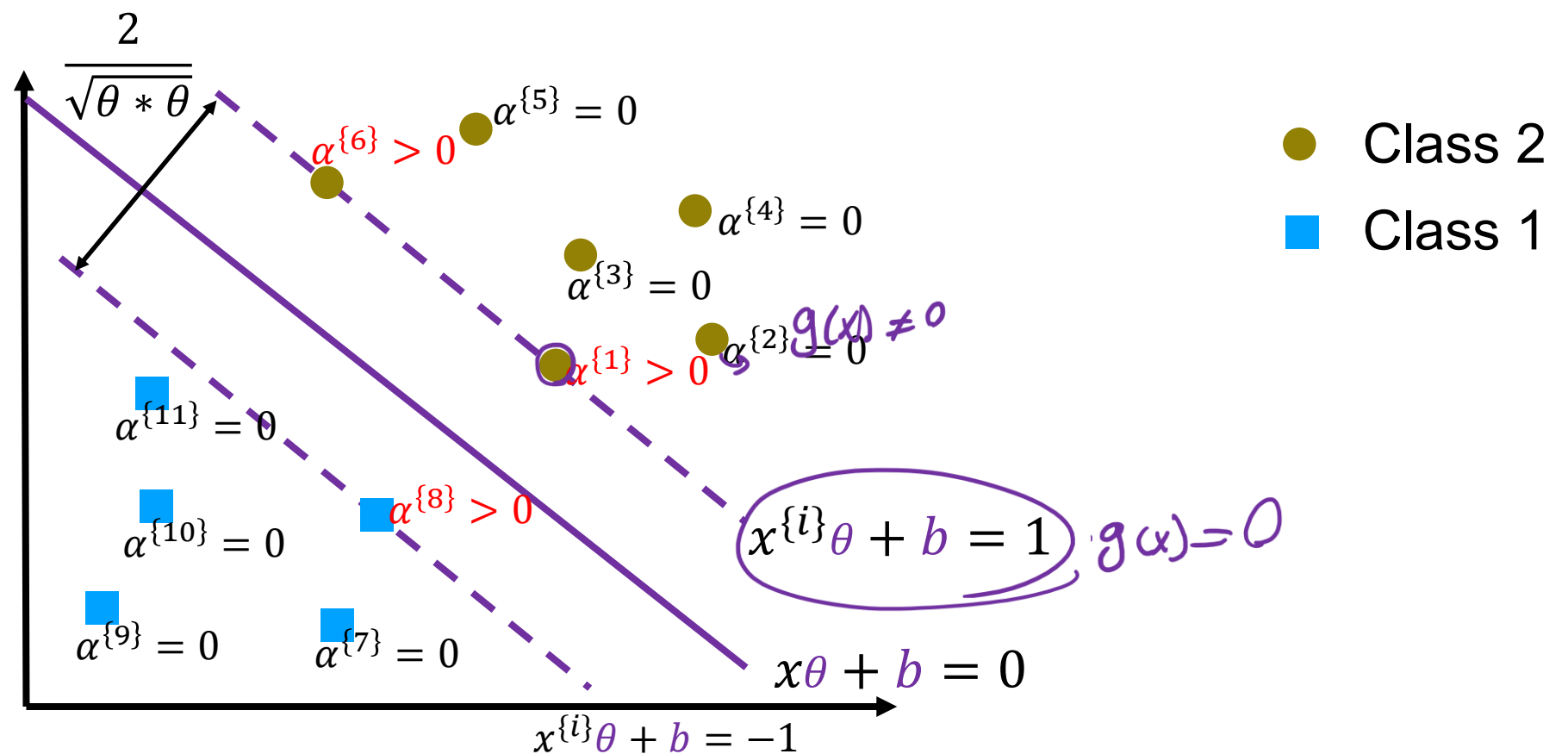
Quadratic programming will give us α

$$\text{Solution: } \alpha = \alpha^{\{1\}}, \dots, \alpha^{\{N\}}$$

$$\text{KKT condition } (\alpha_i g_i(\theta) = 0): \quad \alpha^{\{i\}} (y^{\{i\}} (x^{\{i\}} \theta + b) - 1) = 0$$

$$(y^{\{i\}} (x^{\{i\}} \theta + b) - 1) > 0 \quad \Rightarrow \quad \alpha^{\{i\}} = 0$$

$$(y^{\{i\}} (x^{\{i\}} \theta + b) - 1) = 0 \quad \Rightarrow \quad \alpha^{\{i\}} > 0 \Rightarrow x_i \text{ is a support vector}$$



Training

$$\theta = \sum_{i=1}^N \alpha^{(i)} y^{(i)} x^{(i)}$$

No need to go over all datapoints

$$\rightarrow \theta = \sum_{x^{(i)} \text{ in SV} } \alpha^{(i)} y^{(i)} x^{(i)}$$

and for b pick any support vector and calculate:

$$y^{(i)} (x^{(i)} \theta + b) = 1$$

$$f(x) = x \theta + b > 0 \sim +1$$

$$< 0 \sim -1$$

Testing

For a new test point s

Compute:

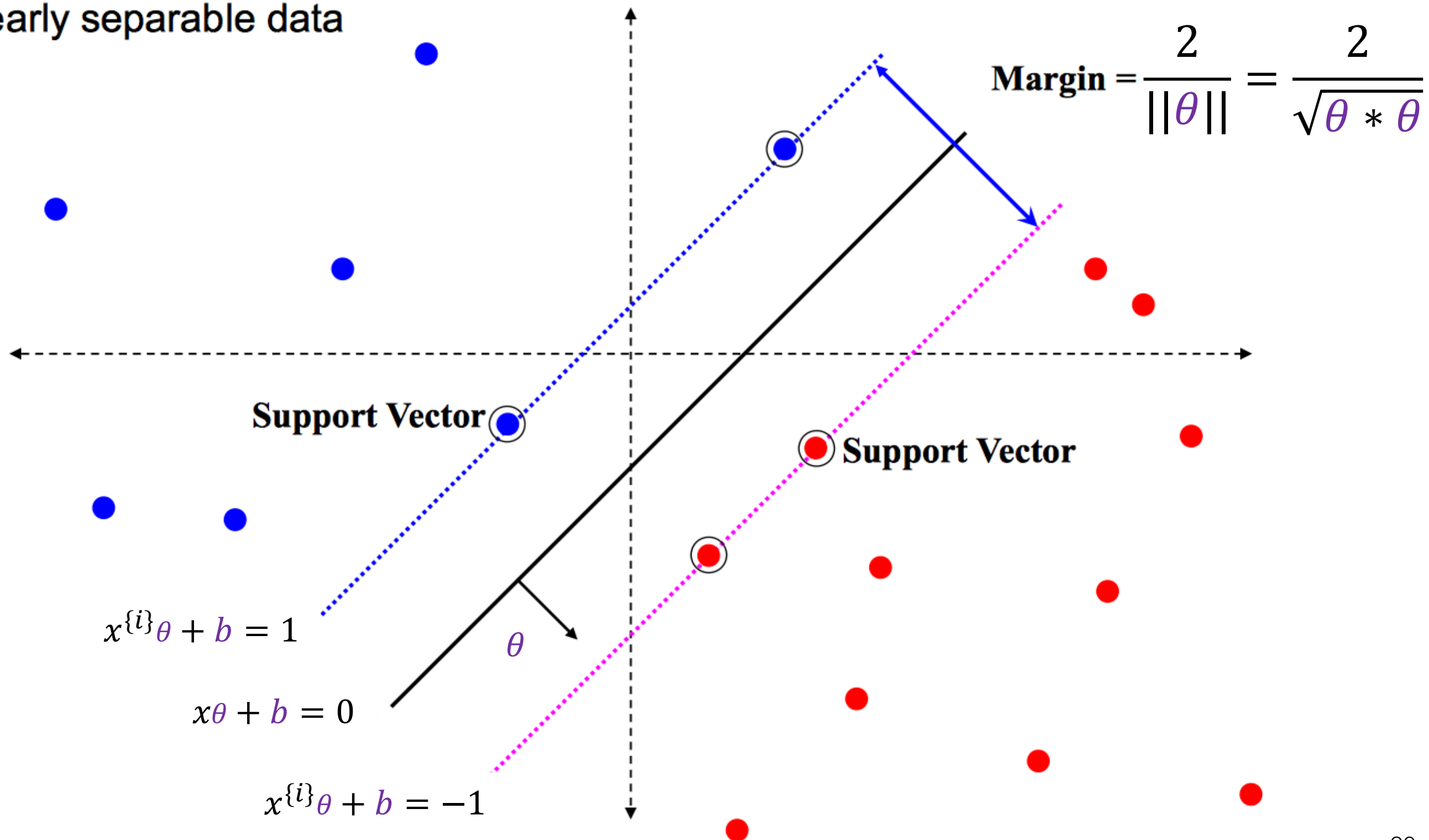
$$\underbrace{s \theta + b}_{\text{LCF primal}} = \sum_{x^{(i)} \text{ in SV} } \alpha^{(i)} y^{(i)} x^{(i)} s^T + b$$

LCF dual form

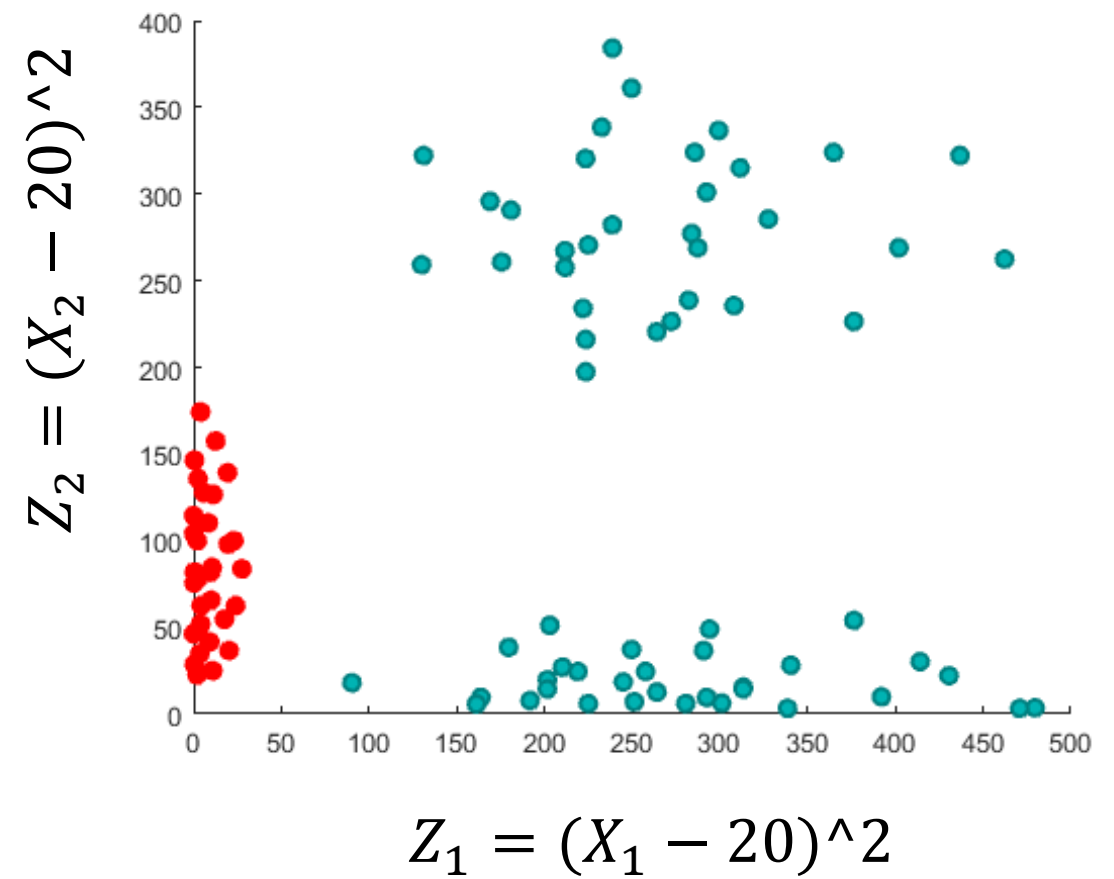
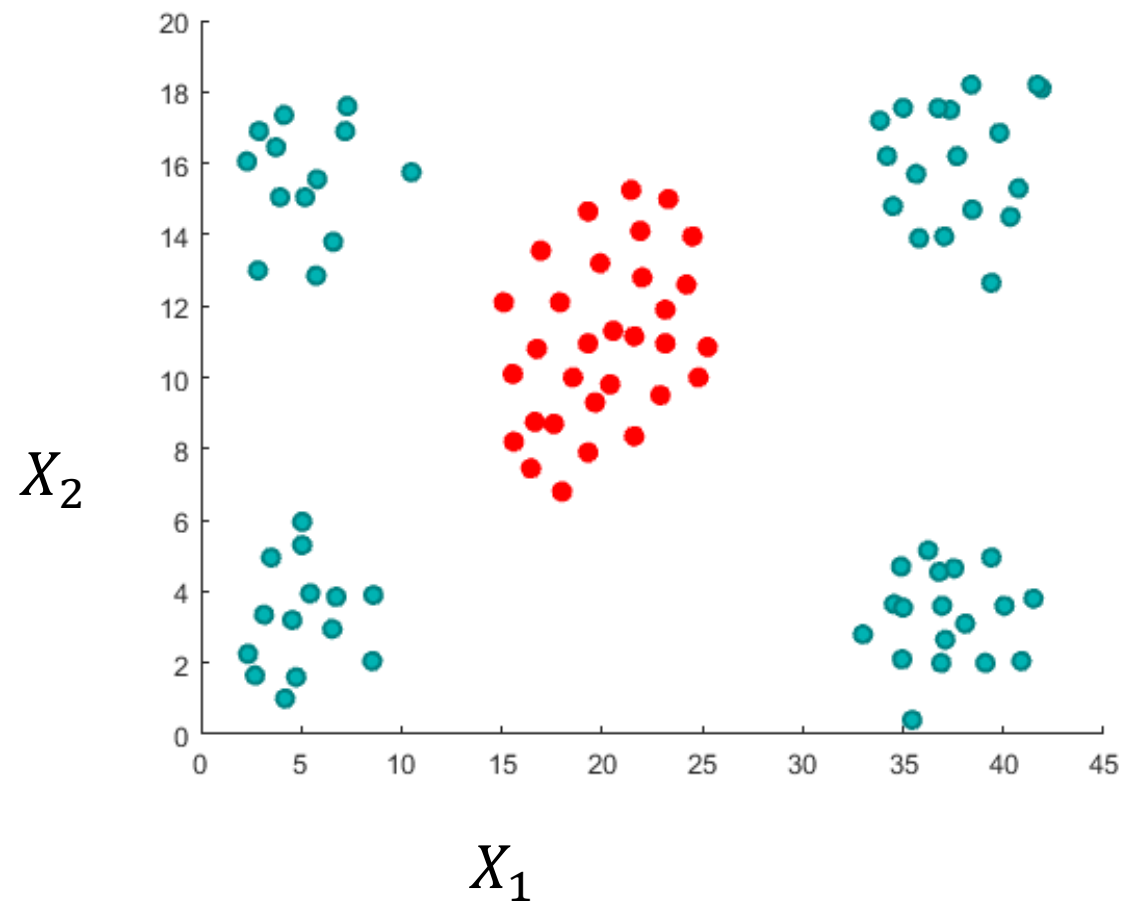
Classify s as class 1 if the result is positive, and class 2 otherwise

Geometric Interpretation

linearly separable data



From x to z space



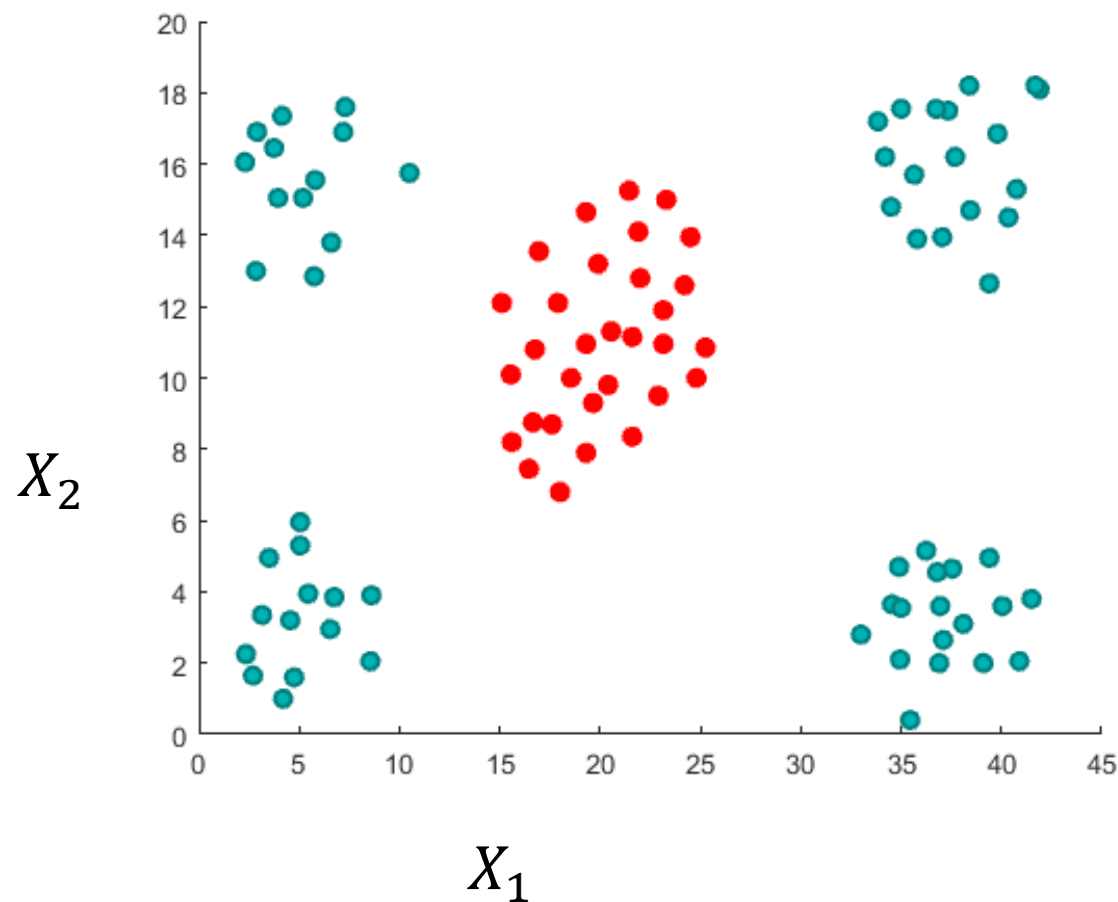
$$X \xrightarrow{(X_1^2, X_2^2)} Z$$

$X_{n \times d}$

In x space

$$\max_{\alpha} \sum_{i=1}^N \alpha^{\{i\}} - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{\{i\}} y^{\{j\}} \alpha^{\{i\}} \alpha^{\{j\}} x^{\{i\}} x^{\{j\}T}$$

$(XX^T)_{N \times N}$

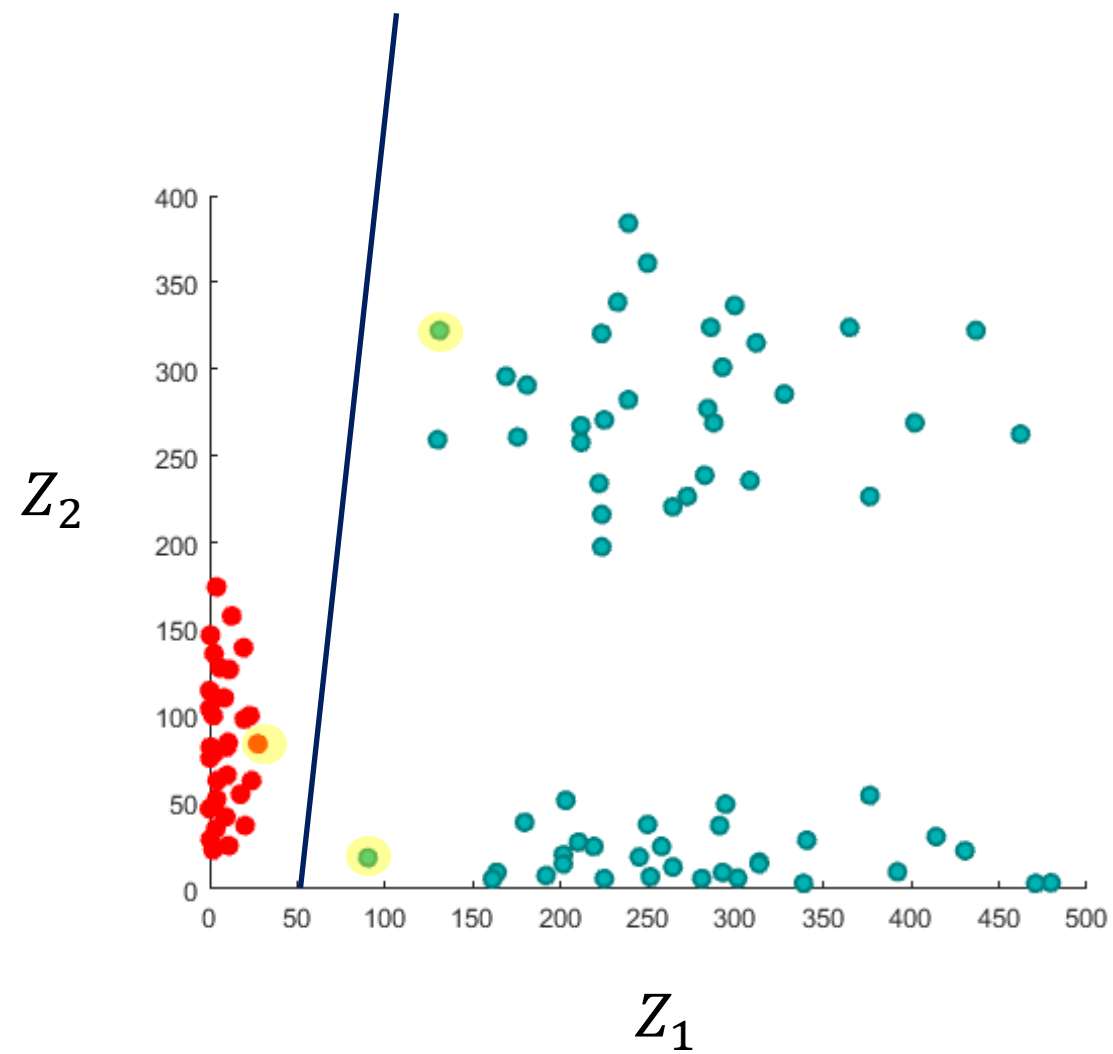


let's say x is $n \times d$
 xx^T will be $n \times n$

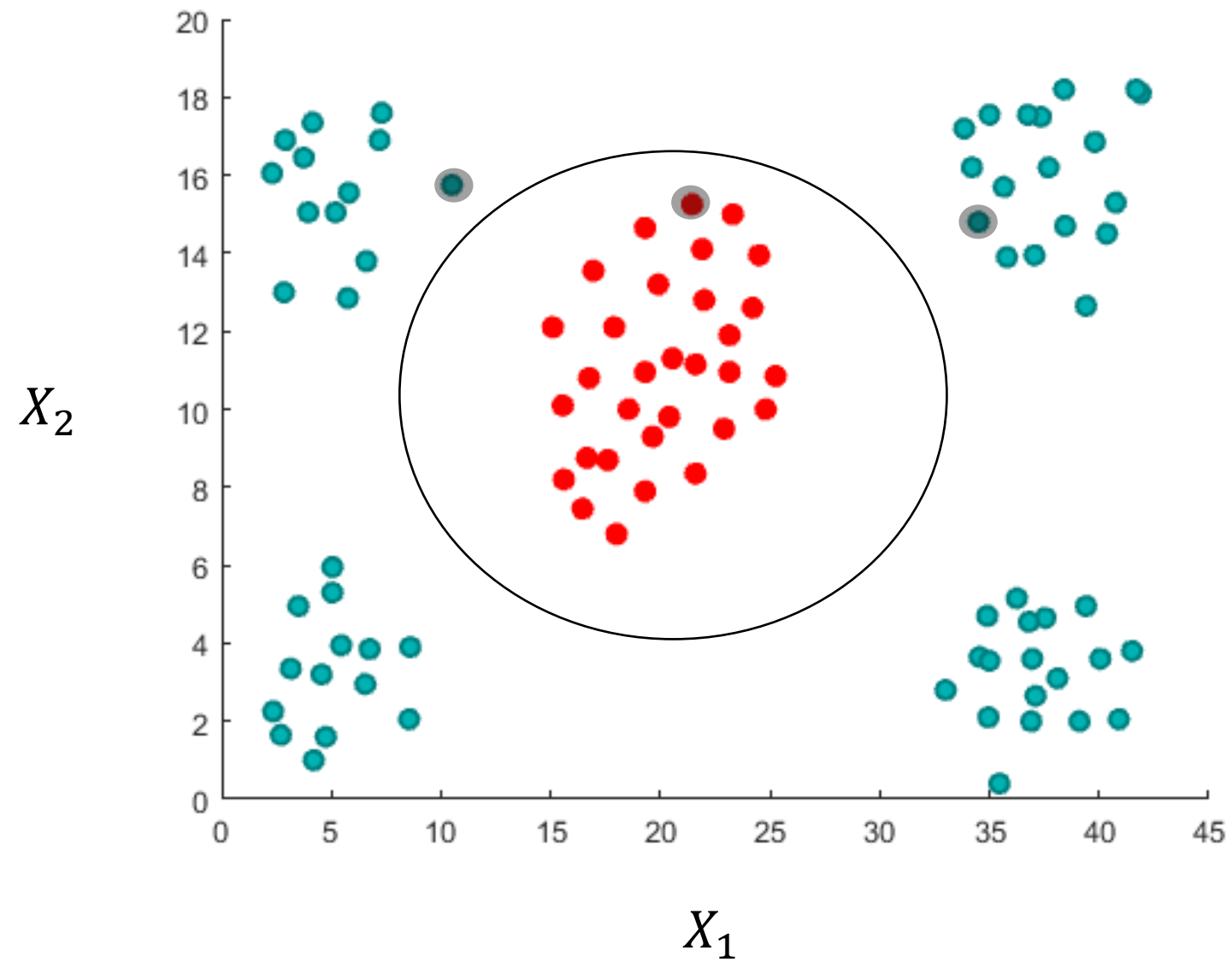
If I add millions of dimensions to x , would it affect the final size of xx^T ?

In z space

$$\max_{\alpha} \sum_{i=1}^N \alpha^{\{i\}} - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N y^{\{i\}} y^{\{j\}} \alpha^{\{i\}} \alpha^{\{j\}} z^{\{i\}} z^{\{j\}T}$$



In x space, they are called pre-images of support vectors



Take-Home Messages

- Linear Separability
- Perceptron
- SVM: Geometric Intuition and Formulation